

UML Case Study Assignment

Smart Library Management System

Name	_____
Roll Number	_____
Course	Object Oriented Analysis and Design
Topic	UML Modelling

Submission order

Questions 1 and 3 contain written answers. Questions 2 and 4-9 are answered using the seven UML diagrams. Every diagram page is labelled with its matching question number.

Question 1 - Requirements and Actors

Identify the functional requirements, non-functional requirements, primary actors and supporting actors. State at least four reasonable assumptions.

Answer

A functional requirement tells us what the system must do. For this library system, the main functional requirements are:

- 1 Users can sign in with their university identity number.
- 2 A user can search the catalogue by title, author, subject or ISBN and check copy availability.
- 3 A librarian can issue and receive a physical book by scanning the user's card and the copy barcode.
- 4 A student can borrow at most five books for fourteen days. A faculty member can borrow at most ten books for thirty days.
- 5 If a title is unavailable, an eligible user can reserve it. The system keeps the requests in a waiting queue.
- 6 When a copy becomes available, the next eligible user is notified. The copy must be collected within two working days.
- 7 A loan can be renewed once if it is not overdue and there is no active reservation for that title.
- 8 The system calculates overdue fines using the currently configured library rules.
- 9 Users can pay a fine online through the payment gateway or at the library desk.
- 10 A separate charge is created when a book is lost or damaged.
- 11 Librarians can add titles, register copies, update copy condition and remove withdrawn copies.
- 12 Administrators can manage user roles, borrowing rules, fine rates and report access.
- 13 The system sends due-date reminders and reservation messages by email or SMS.
- 14 It produces reports for circulation, overdue items, fine collection, popular titles and inventory status.
- 15 Every sensitive action is authenticated and recorded in the audit log.

Non-functional requirements

- **Security:** Only authenticated users can perform protected actions. Access is based on role, and payment or personal data is sent securely.
- **Performance:** Catalogue searches and barcode operations should normally complete within a few seconds.
- **Reliability:** The system must not create two active loans for the same copy. Failed transactions should be rolled back safely.
- **Availability:** The service should be available during university and library working hours. Failure of email, SMS or payment services should not stop normal library work.
- **Usability:** The web and mobile interfaces should be simple, readable and use the same terms.
- **Maintainability:** Loan periods, limits and fine rates should be configurable instead of being fixed in the program.
- **Scalability:** The system should continue to work when the number of users, copies and transactions increases.
- **Backup and recovery:** The relational database should be backed up regularly so that important records can be restored.

Actors

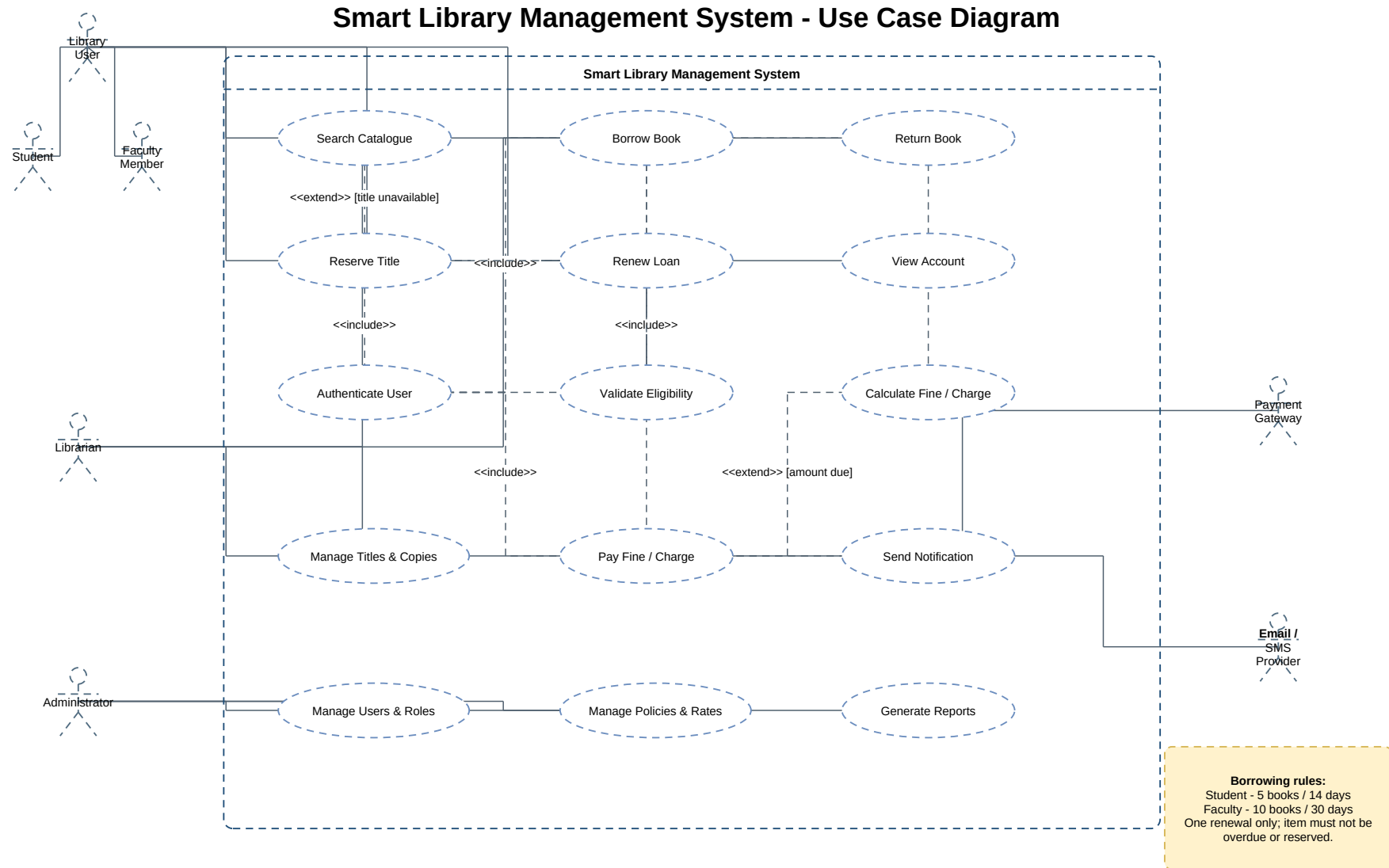
Actor type	Actor	Role in the system
Primary	Student	Searches, borrows, reserves and renews books, and pays fines.
Primary	Faculty Member	Uses the same library services but has a larger limit and longer loan period.
Primary	Librarian	Issues and receives books, manages copies and records desk payments.
Primary	Administrator	Manages roles, policies, fine rates and report access.
Supporting	Payment Gateway	Processes online payments.
Supporting	Email / SMS Provider	Delivers reminders and reservation notifications.
Supporting	Barcode Scanner	Supplies identity-card and book-copy barcode data.

Reasonable assumptions

- 1 Every user has one unique and active university identity number.
- 2 Every physical book copy has its own barcode.
- 3 A blocked or inactive account cannot borrow, renew or reserve a book.
- 4 Only a copy whose status is Available can be issued normally.
- 5 Reservation requests follow first-come, first-served order.
- 6 The two-day collection period is calculated using the library's working calendar.
- 7 A desk payment is marked as paid only after the librarian receives the money.
- 8 If a notification service is unavailable, the message is stored and sent again later.

Question 2 - Use Case Diagram

Prepare a complete use case diagram. Show the system boundary, all actors, major use cases, actor generalization where appropriate, and meaningful include or extend relationships.



Question 3 - Use Case Specification

Write a fully dressed use case specification for Borrow Book. Include preconditions, trigger, main success scenario, alternative flows, postconditions and business rules.

Answer - Fully Dressed Use Case: Borrow Book

Use Case ID	UC-01
Scope	Smart Library Management System
Level	User goal
Primary actor	Librarian
Supporting actors	Library User, Barcode Scanner, User Account, Catalogue / Inventory Service and Notification Service
Goal	Issue one available physical copy to an eligible user and record the correct due date.

Stakeholders and interests

- **Library user:** wants the correct book to be issued quickly and wants to know the due date.
- **Librarian:** wants a simple and accurate issuing process.
- **Library administration:** wants borrowing policies followed and every issue recorded.
- **University:** wants user information and library property to remain secure.

Preconditions

- 1 The librarian is logged in and authorised to issue books.
- 2 The user has a valid university identity card and has brought the physical book to the desk.
- 3 The copy is registered and its barcode can be read.
- 4 The library system and barcode scanner are available.

Trigger

The user asks to borrow a book, and the librarian selects the Borrow Book option.

Main success scenario

- 1 The librarian starts the Borrow Book operation.
- 2 The librarian scans the user's university identity card.
- 3 The system retrieves the user's account and checks that it is active.
- 4 The system checks the user's category, current loans and borrowing limit.
- 5 The librarian scans the barcode of the book copy.
- 6 The system retrieves the Book Title and Book Copy records.
- 7 The system confirms that the copy is available.
- 8 The system checks whether the title has an active reservation. If there is one, it must belong to this user.
- 9 The system selects the correct loan period: fourteen days for a student or thirty days for a faculty member.
- 10 A Loan record is created with the issue date and due date.
- 11 The Book Copy status changes to Issued.
- 12 The system records the action in the audit log.
- 13 An issue confirmation and due date are sent or queued for the user.
- 14 The librarian hands the physical book to the user.

Alternative flows

A1 - Identity cannot be verified

The librarian rescans the card or enters the identity number manually. If verification still fails, no loan is created.

A2 - Account blocked or inactive

The system shows the restriction. The librarian informs the user, and the issue is stopped.

A3 - Borrowing limit reached

If a student already has five loans or a faculty member already has ten, the system rejects the request.

A4 - Unpaid charge prevents borrowing

The user is asked to settle the charge. Borrowing can be tried again after payment.

A5 - Barcode invalid

The librarian scans again or enters it manually. If the copy cannot be identified, it is not issued.

A6 - Copy unavailable

If the copy is already issued, lost, damaged, withdrawn or reserved for someone else, the request is rejected. The user may reserve the title if appropriate.

A7 - Notification service unavailable

The loan remains valid. The confirmation is stored as pending and retried later.

A8 - Failure while saving

The system rolls back the incomplete transaction, leaves the copy status unchanged and shows an error to the librarian.

Postconditions

- **On success:** an active Loan exists, the copy is Issued, the correct due date is stored, the audit entry is written and the notification is sent or queued.
- **On failure:** no incomplete Loan remains, the copy status and active-loan count do not change, and the error is logged.

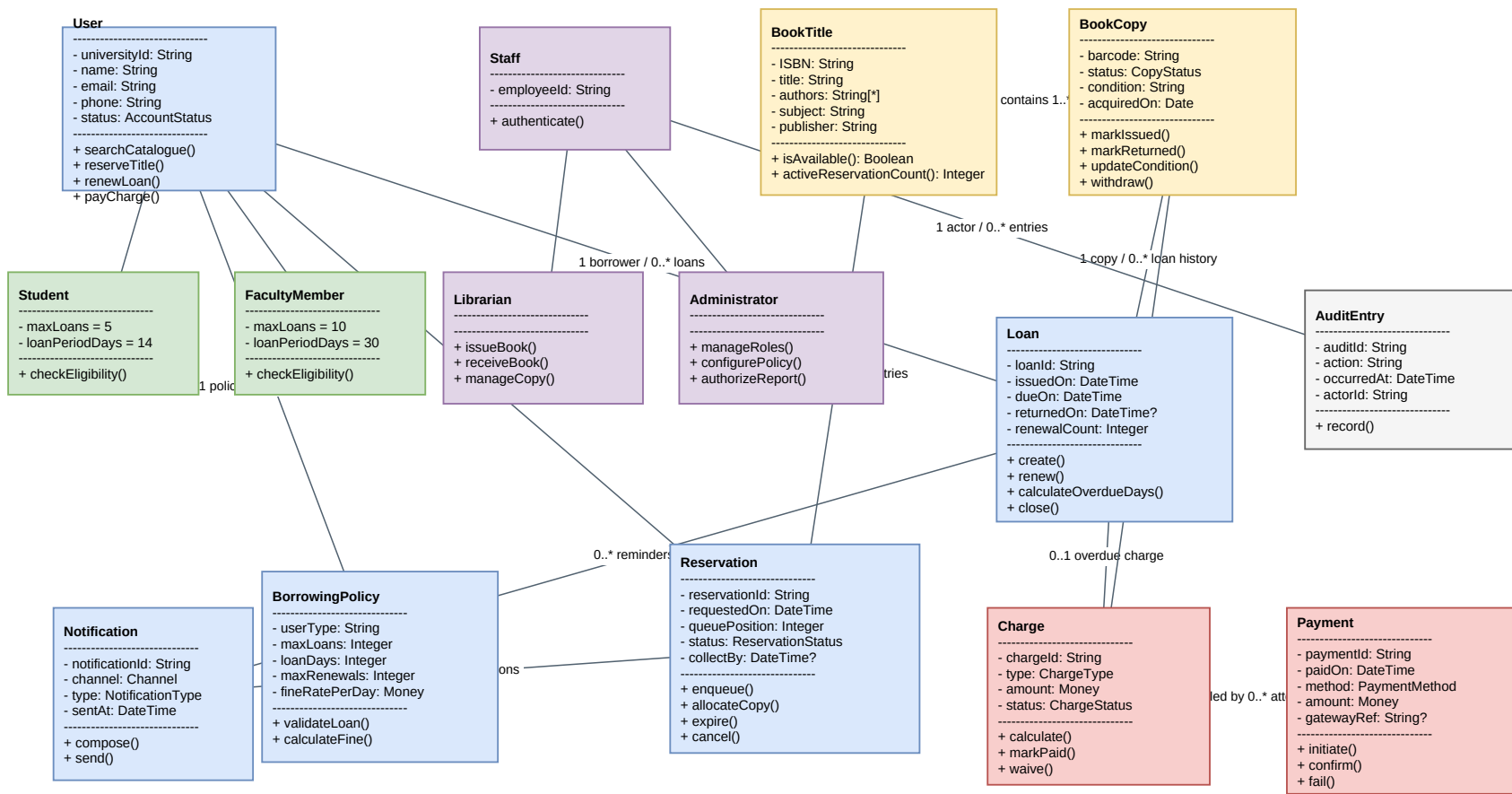
Business rules

- 1 A student may hold at most five books and normally borrows for fourteen days.
- 2 A faculty member may hold at most ten books and normally borrows for thirty days.
- 3 One Book Copy can have only one active Loan at a time.
- 4 A reserved copy can be issued only to the user whose reservation is currently active.
- 5 A reserved copy must be collected within two working days.
- 6 A blocked user is not eligible to borrow.
- 7 Every successful issue is recorded in the audit log.
- 8 The due date is calculated from the policy that is active when the Loan is created.

Question 4 - Domain Class Diagram

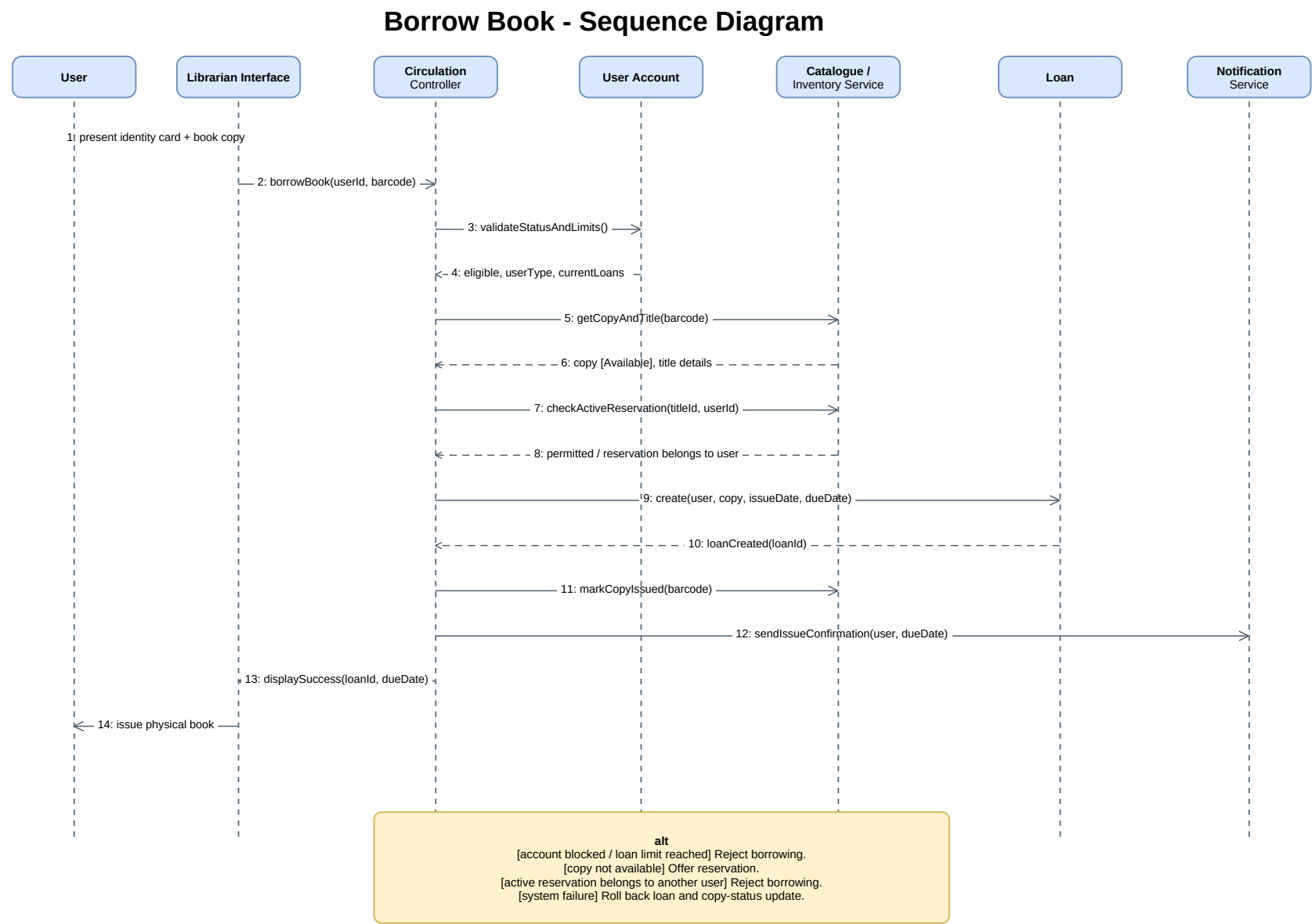
Construct a domain class diagram containing suitable classes, attributes, operations, associations, multiplicities, generalization, aggregation or composition. Distinguish Book Title from Book Copy.

Smart Library Management System - Domain Class Diagram



Question 5 - Sequence Diagram

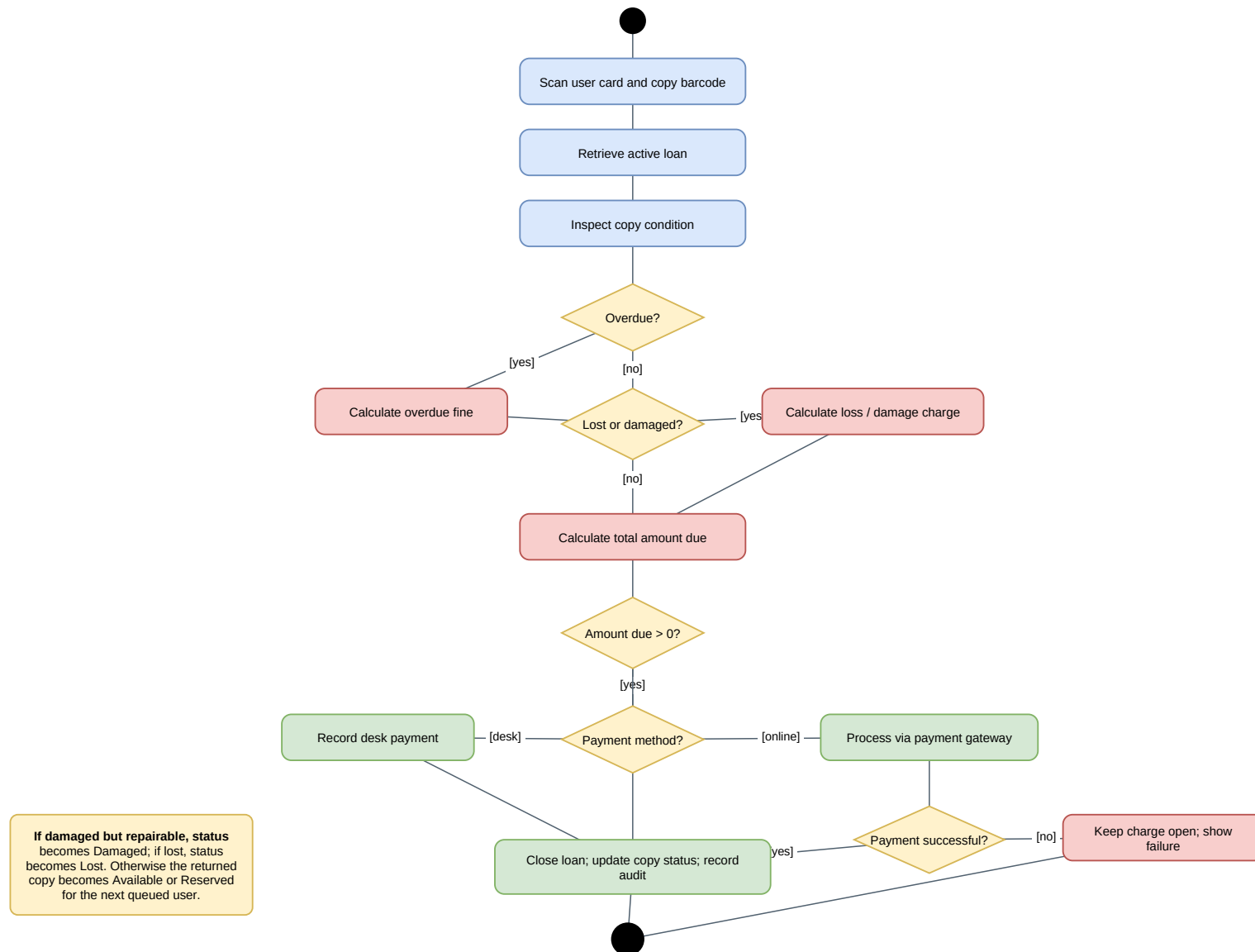
Draw a sequence diagram for Borrow Book. Include the user, librarian interface, circulation controller, user account, catalogue or inventory service, loan object and notification service.



Question 6 - Activity Diagram

Draw an activity diagram for Return Book and Fine Settlement. Include decisions for overdue status, damage or loss, fine calculation, payment method and successful closure.

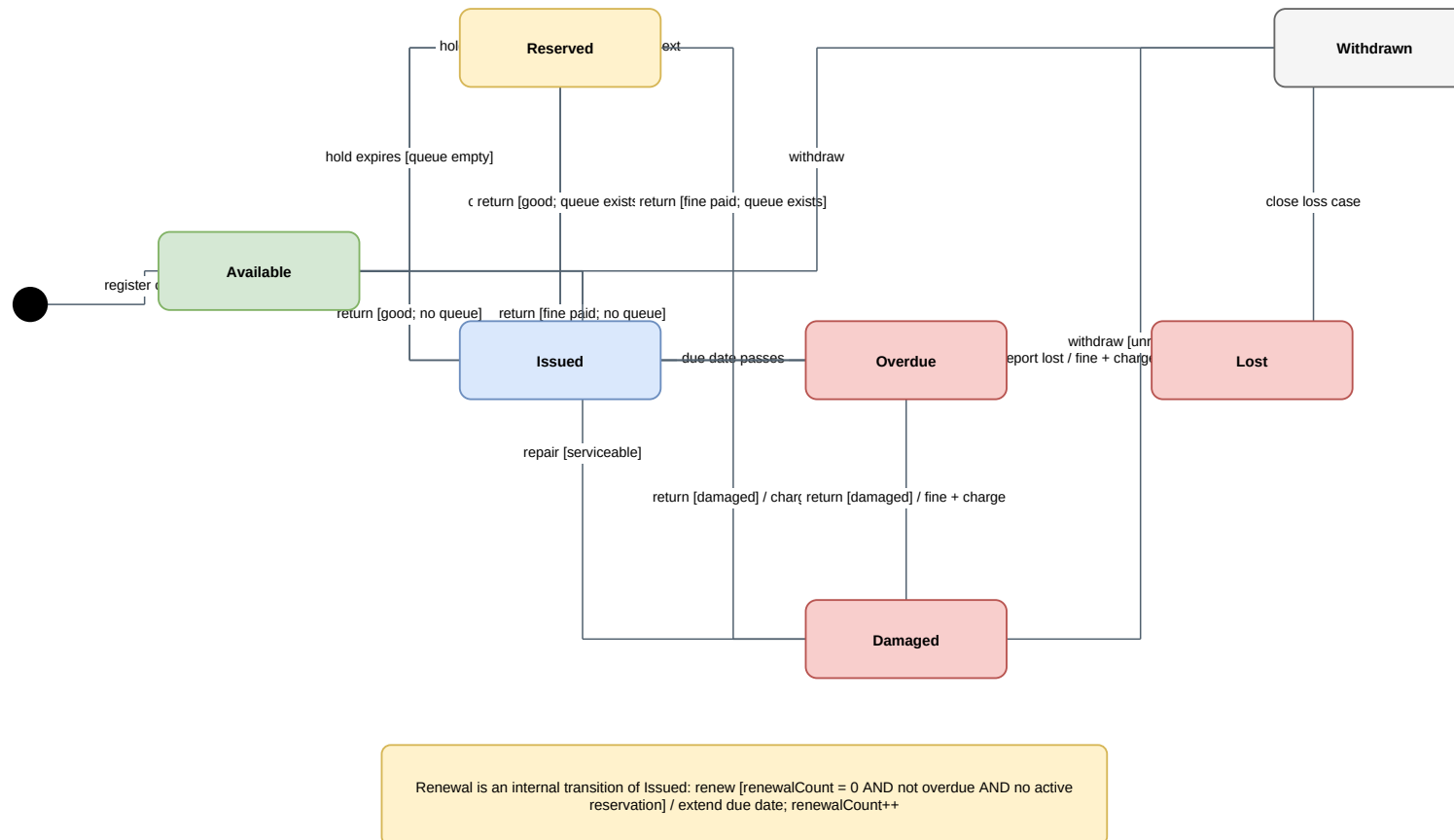
Return Book and Fine Settlement - Activity Diagram



Question 7 - State Machine Diagram

Develop a state machine diagram for Book Copy. Include available, reserved, issued, overdue, lost, damaged and withdrawn states with relevant events and guards.

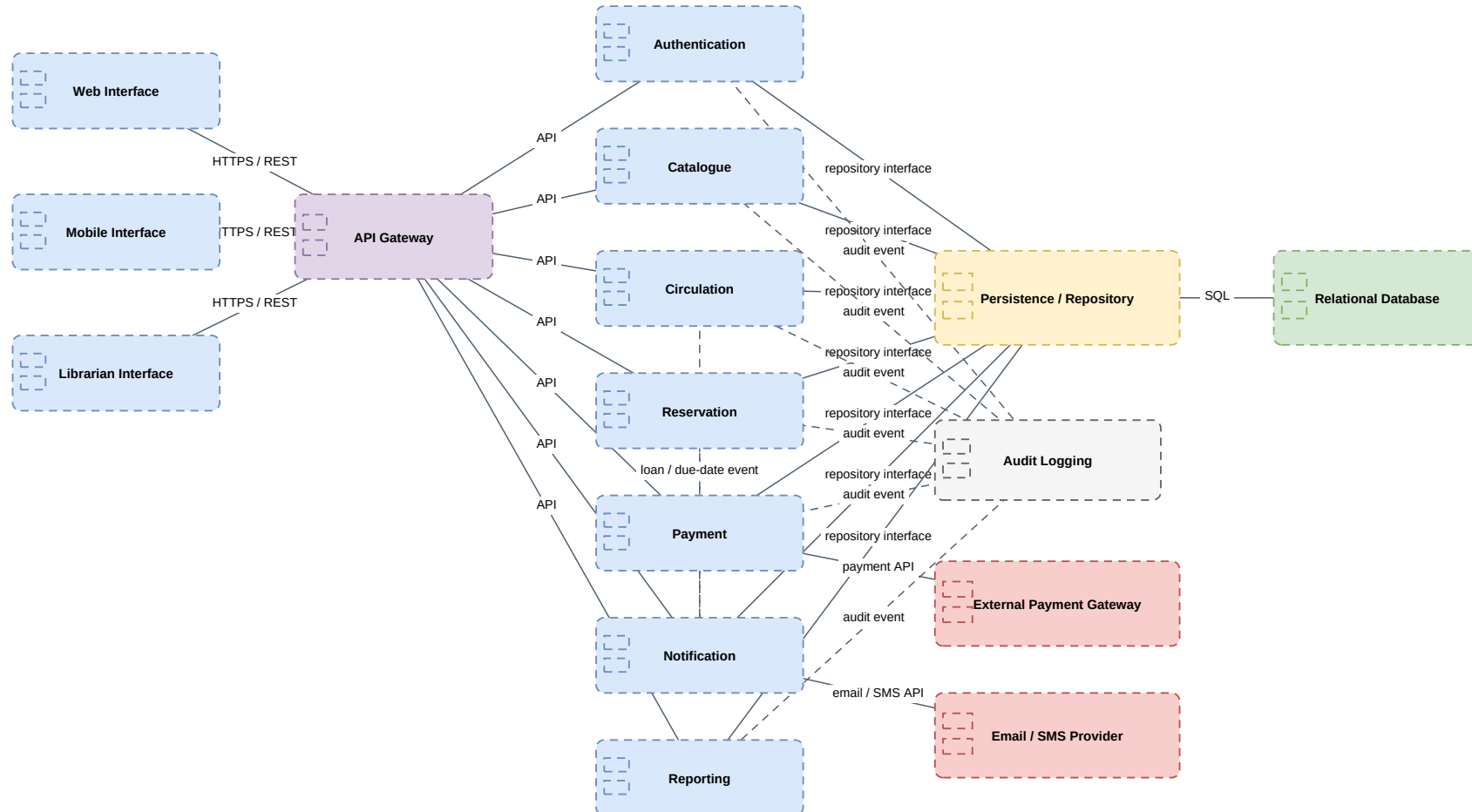
Book Copy - State Machine Diagram



Question 8 - Component Diagram

Prepare a component diagram showing the web or mobile interface, authentication, catalogue, circulation, reservation, payment, notification, reporting and persistence components.

Smart Library Management System - Component Diagram



Question 9 - Deployment Diagram

Prepare a deployment diagram showing client devices, application server, database server, barcode scanner, payment gateway and email or SMS provider.

Smart Library Management System - Deployment Diagram

